Remaining Useful Life Prediction Methodologies With Health Indicator Dependence for Rotating Machinery: A Comprehensive Review

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***Abstract*— Facilitated by artificial intelligence (AI) technologies such as deep learning (DL), predictive maintenance (PdM) has emerged as the most cost-effective maintenance approach for significantly enhancing the safety and availability of industrial equipment. Remaining useful life (RUL) prediction of rotating machinery (RM) is one of the core tasks of PdM, which can assist in maintenance decision-making and optimize asset allocation. Among them, the prognostic approach with health indicator (HI) as an intermediate bridge has attracted increasing attention due to its advantages in combining health condition monitoring (CM) and RUL prediction capabilities. Unfortunately, there has not been a comprehensive review to systematically analyze and summarize the recent advances in HI-dependent prognostic approaches. To fill this gap, this article comprehensively reviews the RUL prediction methods with HI dependence for RM to help researchers quickly outline the current research status and state-of-the-art methods. First, several run-to-failure datasets used to validate the performance of prognostic algorithms are briefly introduced. Second, three types of HI-based prognosis methods for RM, including similarity-based prognostic (SBP) methods, model and data-based statistical methods, and intelligent time series prediction (TSP) methods, are analyzed comprehensively. Finally, the current main challenges and prospects are presented to conclude this article. It is anticipated that the target audience of this article will be practitioners and researchers involved in the field of machinery PdM, providing them with the necessary basic knowledge and technical overview.**

***Index Terms*— Deep learning (DL), health indicator (HI), predictive maintenance (PdM), prognosis, time series prediction (TSP).**

I. Introduction

In THE era of Industry 4.0, the increasing complexity and integration of high-end rotating equipment, such as aero-engines, wind turbines, and high-speed trains, have increased user demands regarding their safety and reliable operation [1], [2]. In order to ensure that mechanical equipment can successfully complete various tasks with low accident rates and maintenance costs, researchers from developed countries in Europe and America began researching maintenance and diagnostic technologies since the 1960s. At the end of the 20th century, prognostics and health management (PHM) technology was born. As shown in Fig. 1, the development of maintenance decision-making strategies for machinery has gone through four stages, from passive reactive maintenance to proactive predictive maintenance (PdM). In reactive maintenance, equipment or components are usually replaced after complete failure [3]. This strategy can take advantage of the maximum life of the equipment and is easy to implement. However, unplanned sudden failures may damage the machine, affect production, and cause casualties. Preventative maintenance with equal time intervals was subsequently adopted by industrial personnel to reduce unplanned downtime and avoid catastrophic accidents [4]. Although the overall malfunction rate of machines has decreased, periodic maintenance has increased planned downtime and may have caused over-maintenance, increasing economic losses. Condition-based maintenance (CBM) can improve the equipment availability and reduce downtime by monitoring a set of critical state parameters of mechatronic equipment [5]. However, it also brings an increase in the cost of condition monitoring (CM) systems. PdM can be regarded as an advancement of CBM, which is characterized by the proactive prediction of the future machinery health status based on the current and historical CM data [6]. With prognostic information, such as remaining useful life (RUL) and failure probability, engineers can schedule an economical maintenance plan before a fault occurs [7]. As a result, PdM typically provides a longer response time for maintenance decisions by utilizing the additional prognostic information compared to CBM.

As critical tasks of PdM, the health CM and RUL prediction of machinery can provide valuable reference information for maintenance decision-making [8], [9]. Health indicators (HIs) can quantitatively evaluate the health status of mechanical equipment and can be combined with some judgment criteria for early fault detection and RUL estimation. In recent years, academia and industry have been widely concerned with the health monitoring and prognostics of rotating machinery (RM) due to the rapid development of artificial intelligence (AI) and advanced sensor technology. After decades of development, the existing RUL prediction methods can be divided into four categories: physical model-based methods [10], model and data-based statistical methods [11], [12], [13], data-driven AI methods [14], [15], [16], and hybrid methods [17], [18], [19].

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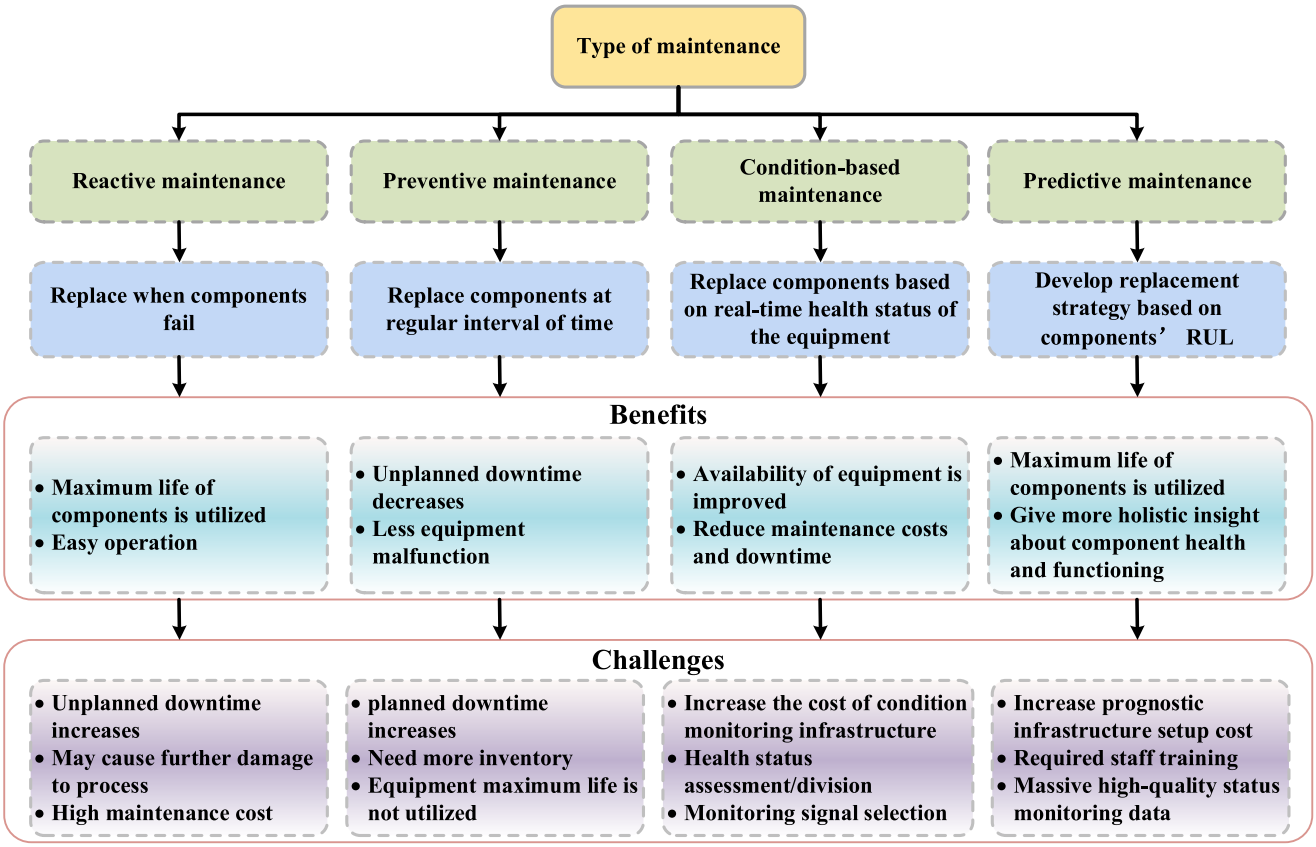


Fig. 1. Maintenance strategies for mechanical equipment with their corresponding benefits and challenges.

Due to the difficulty in establishing accurate physical models for complex mechanical systems, the research work on the first and the last method is relatively limited. Motivated by the Industrial Internet of Things and big data technology, data-driven AI methods have gradually become mainstream and have been widely reported [15], [20]. Specifically, data-driven AI methods have further evolved into two categories: end-to-end pattern recognition and indirect mapping methods with HI dependence. Pattern recognition methods typically utilize AI tools such as extreme learning machine [21], [22], support vector regression [23], convolutional neural networks (CNNs) [24], and transformers [25] to build an intelligent model that directly learns the mapping relationship between the mechanical degradation features and actual RULs. Driven by deep learning (DL) technology, the potential of these methods in addressing the prognostic issues of complex mechatronic systems cannot be ignored. The indirect mapping methods mainly rely on HI as an intermediate bridge, which is achieved by similar pattern matching or step-wise time series prediction (TSP) based on the constructed HI. Furthermore, the model and data-based statistical methods also require the establishment of explicit HI. Therefore, the RUL prediction methods with HI dependence occupy an extremely important position in machinery prognostics. It is important to note that a few scholars have attempted to implement RUL prediction by establishing a mapping relationship between HIs and target RUL labels. Zhang et al. [26] provide case studies using bidirectional long short-term memory (LSTM) networks, time-delay neural networks, and prediction functions as mapping models. However, this type of approach exhibits weak dependence on HI and can be regarded as an evolution of pattern recognition methods.

In the literature investigated, some review articles have contributed to introducing the prognostic approaches for RM. For example, Si et al. [11] conducted a detailed review of statistical data-driven RUL prediction methods, systematically elaborating on their basic theories and critically discussing their pros and cons. Kan et al. [27] reviewed prognostic techniques for RM operating under nonlinear and nonstationary conditions, discussing their general concepts and application scenarios. El-Thalji et al. [28] presented predictive health monitoring methods for rolling element bearings up to 2014 and briefly reviewed three types of prognostic analysis methods. In [1] and [29], RUL estimation methods for critical components of wind turbines are summarized to help reduce wind farm operation and maintenance costs. For the particle filter (PF)-based machinery prognosis method, Jouin et al. [30] introduced its theoretical basis and reviewed the applications of different filters in prognostics up to 2015. Tahan et al. [31] and Hanachi et al. [32] analyzed the health CM and RUL prediction techniques for gas turbines. In 2017, Lei et al. [33] provided a systematic review of machinery health prognostics from data acquisition to RUL prediction. In addition, a comprehensive investigation and discussion on vibration-based gear health monitoring and prognosis methods were conducted [2], [34].

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The studies mentioned above provide valuable insights into the health prognostic techniques of RM from different perspectives, such as from the perspective of the object, methodology, or prognostic procedure. However, they still have some limitations.

1) Although numerous machinery prognostic methods have been discussed in the existing review papers, RUL prediction methodologies with HI dependence have not been comprehensively summarized and analyzed. In particular, the prognosis methods based on TSP of HI have not been systematically reviewed, thereby failing to provide a constructive reference for the relevant practitioners.

2) CM data acquisition is an important preliminary step in machine health monitoring and RUL prediction. Nevertheless, the valuable public datasets in this field and their characteristics have not been thoroughly introduced and analyzed.

3) Most reviews were published before 2018 [11], [27], [28], [29], [30], [31], [32], [33]. However, a huge amount of relevant research has been published in recent years, while the earlier reviews did not cover the latest progress in this field.

HI-dependent methods first construct an HI to quantitatively assess the equipment's current operating status. This HI is then used to predict the RUL of the mechanical equipment. Compared to using raw sensor data directly, this approach simplifies the model and enhances the understanding of the degradation behavior. Moreover, these methods can leverage the constructed HI to detect early signs of degradation and capture the degradation trend, enabling more proactive and informed maintenance decisions. To summarize, it is necessary to conduct a comprehensive review of the latest advances in RUL prediction methodologies with HI dependence for RM. This article fills the above gaps, and its main contributions are summarized as follows.

1) This article comprehensively covers three HI-dependent prognostic approaches for RM, namely, model and data-based statistical methods, similarity-based prognostic (SBP) methods, and TSP ones. It also provides a systematic perspective for concerned practitioners and novice researchers.

2) The publicly available datasets for the full life cycle of RM are summarized in this article, which can help researchers quickly find suitable datasets to validate the effectiveness of their algorithms.

3) Several core points of SBP methods are discussed in detail, including similarity measure criteria, similar instance matching rules, and integration of prediction results.

4) The RM prognosis methods based on TSP of HI are summarized systematically for the first time, providing new ideas for solving the prognostic problems under the limited samples in actual industry.

The rest of this article is arranged as follows. Section II reviews the public life-cycle datasets and the HI construction methods. In Section III, the prediction framework of similarity-based methods is summarized. In Section IV, the overview of TSP-based methods is reviewed. The challenges and prospects are written in Section V. Finally, Section VI concludes this article.

II. Data Acquisition for Rotating Machinery

A. Condition Monitoring Data Acquisition for Rotating Machinery

CM data acquisition is the first key step in implementing RUL prediction for RM. Monitoring signals containing rich status information can be obtained by installing different types of sensors, such as accelerometers, acoustic emission, and temperature sensors, at appropriate locations on the monitored equipment. Unfortunately, CM data in real industrial environments are difficult to obtain because safety-critical equipment is generally not allowed to operate to failure, coupled with long acquisition cycles and harsh operating environments. Therefore, most literature has chosen simulated or accelerated degradation experimental datasets for validation, while a few scholars have applied datasets from real industrial scenarios to develop prognostic methods. However, no public industry dataset is currently available due to trade secrets.

In order to help researchers easily find suitable datasets for developing and validating prognostic algorithms, this article summarized some publicly available full life-cycle datasets. Table I lists the key information of several typical RM prognostic datasets, including monitored objects, operating conditions, failure modes, and monitoring variables. More detailed information can be obtained from the provided references, such as sample size, sampling frequency, and RUL labels. Due to the different characteristics of various datasets, they can prompt researchers to propose innovative prognostic algorithms to estimate the RUL. The C-MAPSS dataset is widely used to develop and validate various data-driven RUL prediction methods due to a large number of life-cycle samples. Meanwhile, various sensors allow researchers to develop effective feature fusion algorithms. Moreover, this dataset includes six different operational conditions. The time-varying operating condition leads to the fluctuation of degradation trend. Thus, it is also suitable for studying the relationship between the time-varying operational condition and degradation trend. However, since this dataset is a simulation one, it has certain limitations in reflecting the real degradation process of turbofan engines. Milling dataset was collected by using a real milling machine instead of simulation models or experimental platforms, so it is closer to industrial applications. This dataset has different kinds of signals including acoustic emission signals, vibration signals, and electric current signals. Therefore, this dataset is also applicable for researching the RUL prediction based on multisensor fusion. IMS dataset, FEMTO-ST dataset, and XJTU-SY dataset were all acquired from the accelerated degradation tests of bearings. These bearings were naturally degraded without being seeded a fault in advance. Different from the FEMTO-ST dataset, IMS and XJTU-SY datasets provide the fault pattern of each bearing, which can provide valuable information for researchers to study the relationship between different fault patterns and their degradation trends. The CQU dataset was acquired under seven different operating conditions via two gear contact fatigue testing machines, and it contains two typical types of gear faults. However, this dataset only contains vibration signals, and the working condition is not time-varying during a

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TABLE I
TYPICAL PUBLICLY AVAILABLE DATASETS FOR THE PROGNOSIS OF ROTATING MACHINERY

Name	Monitored object	Release institution/Year	Property	Operating conditions (OC)	Failure modes	Monitoring variables	Download link
C-MAPSS dataset [35]	Turbofan engine	NASA Ames Prognostics Center of Excellence/2008	Simulation dataset	Single OC: subset #1, #3. Six OCs: subset #2, #4.	HPC degradation: subset #1, #2. HPC and fan degradation: subset #3, #4.	21 sensor variables: temperature, pressure, speed, bypass ratio, bleed enthalpy, etc.	
Milling dataset [36]	Milling cutter	UC Berkeley BEST Lab/2007	Experimental dataset	Four sets of milling parameters are used for two different materials. (8 OCs)	Flank wear exceeded the preset failure threshold.	Acoustic emission, vibration, current signals.	https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/
IMS bearing degradation dataset [37]	Bearing	Center for IMS, University of Cincinnati/2007	Experimental dataset	Single OC: 2000rpm/6000 lbs load.	Inner race defect, roller element defect, outer race failure.	Vibration signals	
FEMTO-ST bearing degradation dataset [38]	Bearing	FEMTO-ST institute/2012	Experimental dataset	Three OCs: 1800rpm/4kN, 1650rpm/4.2kN, 1500rpm/5kN.	Unavailable	Vibration and temperature signals	
XJTU-SY bearing degradation dataset [39]	Bearing	Xi'an Jiaotong University & Changxing Sumyoung Technology Co., Ltd/2019	Experimental dataset	Three OCs: 2100rpm/12kN, 2250rpm/11kN, 2400rpm/10kN.	Inner race, out race, cage, roller element faults and compound faults.	Vibration signals	https://www.mediafire.com/folder/m3sij67rizpb4/XJTU
CQU gear degradation dataset [40]	Gear	SKLMT of Chongqing University/2024	Experimental dataset	Seven OCs: 500rpm/1.4kN·m, 1000rpm/1.3kN·m, 1100rpm/4.4 kN·m, ⋮	Gear pitting, tooth breakage.	Vibration signals	https://qinyi-team.github.io

life-cycle test. Therefore, this dataset can be applied to validate the transfer RUL prediction tasks across constant working conditions, cross-machine RUL prediction tasks, and federated transfer RUL prediction tasks.

Since most of these datasets were acquired under the experimental environments, they are less affected by noise. To investigate the impact of noise on the RUL prediction results, artificial noise can be added to the sampled signals. Moreover, by deleting some data from the life-cycle data, the incomplete dataset can be generated to verify the RUL prediction methods. Despite the efforts of academics in benchmark dataset sharing, the complexity of degradation process in experimental datasets is far from representative of real-world industrial issues. Therefore, more realistic prognostic datasets are needed to improve the models' application capabilities in the future.

B. Health Indicator Construction for Rotating Machinery

After acquiring the CM data of the RM, an HI that adequately characterizes its degradation process needs to be constructed. As the intermediate bridge, HI can not only be used to quantitatively assess the equipment's current operating status but also lay the foundation for the subsequent RUL prediction. Since this article focuses on the discussion of HI-dependent prognosis methods, only a brief summary of HI construction methods is presented as follows.

According to the existing literature, the HI construction methods can be mainly categorized into three types: signal processing (SP)-based, traditional machine learning (ML)-based, and DL-based HI, as illustrated in Fig. 2, and their advantages and disadvantages are shown in Table II. The first type of HI is usually obtained based on various statistical parameters of CM data in the time domain, frequency domain, and time–frequency domain, which are obtained by SP techniques such as filtering, time synchronous averaging, Fourier transform, short-time Fourier transform, wavelet transform, and empirical mode decomposition [41]. Last but not least, the entropy-based HI construction methods have also received attention, as the mechanical vibration signals during the deterioration probably have increased signal uncertainty. SP-based HIs are characterized by their clear physical meanings. However, their construction process necessitates specific physical knowledge and relies heavily on expert experience. Modern mechanical equipment is becoming increasingly complex, and the volume of collected data is growing exponentially. These factors significantly limit the applicability of SP-based methods.

Compared with SP-based HI, ML-based approaches break the limitation of understanding fault mechanisms and responses [42]. Specifically, the ML-based method can construct HI from two aspects: multifeature fusion and similarity distance measurement of different operating states. The former can be achieved by the varied dimensionality reduction algorithms, such as principal component analysis (PCA), isometric mapping, and locality preserving projection, thereby effectively avoiding the loss of state information caused by a single physical HI. These methods are relatively easy to

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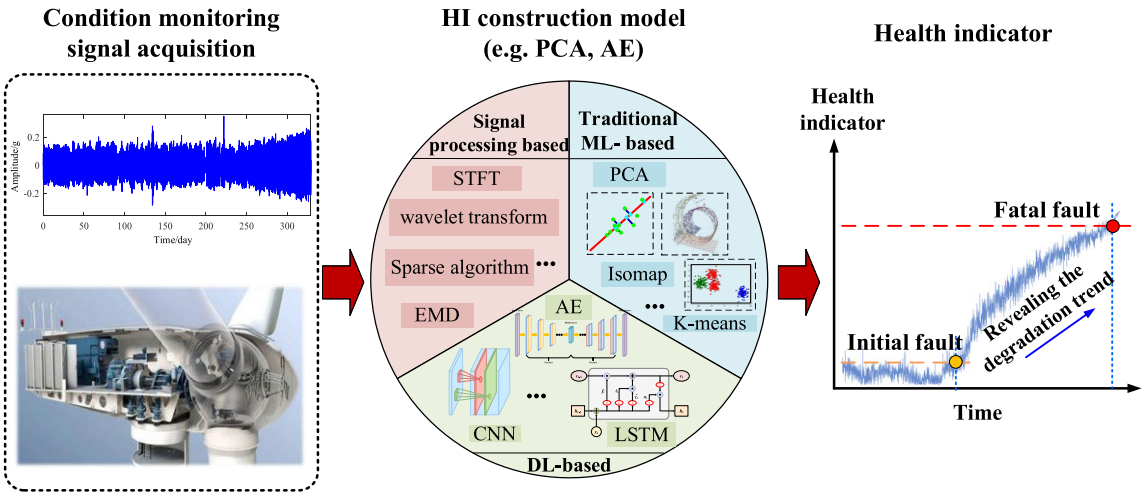


Fig. 2. Overview diagram of various HI construction methods for rotating machinery.

TABLE II
DIFFERENT HI CONSTRUCTION METHODS

Categories	Advantages	Disadvantages	Applicable conditions
SP-based HIs	Specific physical meaning; Easily interpretable; High sensitivity	Provide some specific information about the equipment structure and operation conditions; Poor robustness; Experience and expertise;	Good data acquisition conditions; Good data quality;
ML-based HIs	No reliance on expert knowledge and specific device information; Well-adapted;	Lack of explainability; Hand-crafted features; Information loss; Prone to overfitting;	Limited data samples; Requirement for rapid deployment;
DL-based HIs	Automatic learning feature; High accuracy; Good generalization ability;	Complex models; Lack of explainability; Requirements for computing resources and storage resources; Longer training time;	Large-scale data; Complex patterns and correlations in the data;

implement. Recently, the research on efficient nonlinear fusion methods has become mainstream. However, the quality of the synthesized HI is strongly correlated with the screened degradation features. The similarity measurement methods primarily rely on the deviation between degradation data and healthy samples. On the one hand, this type of method can utilize such metrics as Euclidean distance, Mahalanobis distance, and Kullback–Leibler (KL) divergence to measure the similarity of data or distributions, hence directly obtaining the deviation between the monitoring signal and the reference baseline. On the other hand, these algorithms, such as K-means, fuzzy C-means, and Markov models, can be used to learn the latent distribution space from normal samples and then calculate the HI indirectly. ML-based HIs, despite not requiring specific physical knowledge to guide the construction process, are highly dependent on feature engineering. Moreover, they are not well suited for handling large-scale monitoring data scenarios, which are common in modern industrial applications.

The last type of HI construction approach originates from the rise of DL techniques, including both supervised and unsupervised implementation pathways. The unsupervised construction approach is similar to the ML-based one [43], which is also based on feature fusion and similarity measurement. Among the unsupervised DL-based approaches, the encoder–decoder structure in autoencoder (AE) is extensively applied, benefiting from its advantages in information compression and reconstruction. In feature fusion scenarios, the intermediate embedding layer of AE is usually dimensionally reduced to obtain a 1-D HI. Furthermore, various reconstruction errors based on healthy benchmarks are commonly used strategies to measure the similarity. In contrast to the former, the supervised approach utilizes the labels from the degradation model to constrain the degradation curve and generate HI [43], [44]. So far, the commonly used degradation models include the linear function model, quadratic function model, power function model, and exponential model. Then, the HI with good monotonicity and robustness can be obtained by selecting a suitable degradation model based on the characteristics of CM signals. Moreover, the quality of HI is affected by the degradation feature extractors, such as CNN, LSTM, and other deep neural networks. DL-based methods have the capability to automatically extract hidden features and complex nonlinear relationships from large-scale monitoring data. However, these methods suffer from a lack of interpretability, making it difficult to understand the underlying physical meaning of the constructed HIs. Additionally, they demand substantial computing resources, which can be a challenge in resource-constrained environments.

HI directly impacts the accuracy and effectiveness of the machine RUL prediction model. An appropriate HI can better reflect the health status and degradation tendency of machine, thereby enhancing the predictive capability of model. Therefore, the selection of HI is an essential step to ensure the accuracy and effectiveness of RUL prediction model. Currently, the HI evaluation metrics mainly include monotonicity, correlation, robustness, and hybrid metrics. In order to calculate these evaluation metrics, the constructed HI vector needs

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to be decomposed into an average degradation trend and a residual term, which can be represented as

$$X(t_k) = X_T(t_k) + X_R(t_k) \tag{1}$$

where $X(t_k)$ is the HI value at time t_k, $X_T(t_k)$ denotes the average degradation trend at time t_k, and $X_R(t_k)$ represents the residual term at time t_k. The average degradation trend can be obtained by smoothing algorithms, such as polynomial fitting, exponential smoothing method, and wavelet denoising.

Without maintenance, the degradation of mechanical equipment is irreversible. Thus, the constructed HI should have a uniform upward or downward degradation trend. The commonly used calculation formula of monotonicity is written as

$$\text{Mon}(\mathbf{X}) = \frac{1}{K-1}\left|\sum_K S(X_T(t_{k+1}) - X_T(t_k))\right| \tag{2}$$

where K is the total number of HI series and $S(\bullet)$ is the sign function.

The HI should exhibit a clear degradation trend strongly aligned with the running time, reflecting the heightened likelihood of performance decline with the running time. The calculation formula of correlation is often written as

$$\text{Corr}(\mathbf{X}, \mathbf{T}) = \frac{\left|K\sum_K X_T(t_k)t_k - \sum_K X_T(t_k)\sum_K t_k\right|}{\sqrt{\left[K\sum_K X_T(t_k)^2 - \left(\sum_K X_T(t_k)\right)^2\right]\left[K\sum_K t_k^2 - \left(\sum_K t_k\right)^2\right]}}. \tag{3}$$

Measurement noise and unstable/variable working conditions introduce variability into HI values, potentially obscuring the true degradation trend and leading to unreliable RUL prediction. A suitable HI can minimize these influences and exhibit a smooth degradation trend. This property is defined as robustness, which can be calculated by

$$\text{Rob}_1(\mathbf{X}) = \frac{1}{K}\sum_K \exp\left(-\left|\frac{X_R(t_k)}{X(t_k)}\right|\right). \tag{4}$$

Considering that a single HI evaluation metric is insufficient for selecting a suitable HI, a hybrid metric is developed by integrating multiple metrics. Its calculation formula is written as

$$\text{HM}(\mathbf{X}) = \omega_1\text{Mon}(\mathbf{X}) + \omega_2\text{Corr}(\mathbf{X}) + \omega_3\text{Rob}(\mathbf{X}) \tag{5}$$

where ω is utilized to flexibly adjust the proportion of various properties, which satisfies $\sum_{i=1}^{3}\omega_i = 1$ and $\omega_i \geq 0$. In addition to the above calculation formulas, other improved evaluation metrics can be referred to [59]. Researchers can flexibly choose the aforementioned evaluation metrics for HI selection based on the usage scenario and needs.

III. Similarity-Based Prognostic Methods

A. Overview of Similarity-Based Methods

Compared with RUL prediction methods based on pattern recognition, the similarity-based prediction methods are less researched. However, it is more intuitive and easier to interpret. In the field of prognostics, fuzzy instance models and prediction methods based on similarity match matrix (MM) are considered the prototype of SBP methods [45], [46]. Wang et al. [47] optimized the similarity matching and achieved the best score in the IEEE PHM 2008 Challenge, marking the mature application of the SBP method. The implementation of SBP approach is generally based on the following assumptions.

1) Run-to-failure data from multiple identical or similar mechanical equipment can be obtained, where CM data can be multisource heterogeneous.

2) In-service equipment (i.e., test instances) and historically failed equipment (i.e., training instances) share similar degradation patterns and fault types under the similar operating conditions.

3) The historical failure instances recorded in the HI curve library contain typical degradation paths of mechanical equipment, which are allowed to infer the unknown degradation paths of test instances.

4) The degradation process of different equipment can start from various initial deterioration degrees. Moreover, this prediction method does not consider sudden catastrophic failures.

Therefore, the SBP method has unique advantages. It does not need to set the failure threshold (FT), so it can easily achieve the online prediction, and failure instances can be easily incorporated into the HI curve library. However, a certain number of failure instances are necessary, and the number of reference instances affects the accuracy of SBP method. Engineers can reasonably choose this method based on the actual application scenarios.

The trajectory similarity-based prediction (TSBP) method was systematically expounded in [48], which mainly consists of four parts: data acquisition, HI curve library construction, curve similarity matching, and prediction results integration, as shown in Fig. 3. Different from the pattern recognition prediction method, SBP method requires explicit extraction of low-dimensional representation vectors (i.e., HIs) of each failure instance, forming a reference HI curve library. Each unique HI trajectory characterizes a possible degradation mode of the mechanical system. Since CM data acquisition and HI construction have been discussed in Section II, this section will focus on the last two steps of the SBP approach.

B. Curve Similarity Matching

In the second step of the SBP method, an HI curve library $\mathbf{\Omega} = \left\{\boldsymbol{H}^j\right\}_{j=1}^{J}$ of historical failure instances is constructed, where $\boldsymbol{H}^j = \left[h_1^j, h_2^j, \ldots, h_{T_j}^j\right]^T$ represents the HI curve of the jth failure instance in the library. For the current moment T_c, the curve similarity matching procedure aims to find the first few most similar future degradation paths with a known degradation trajectory $\boldsymbol{H}^{\text{test}} = \left[h_1^t, h_2^t, \ldots, h_{T_c}^t\right]^T$ of a test device through sequential search. Similarity matching is the core of the SBP method. Moreover, two key factors need to be carefully considered in this process: similarity measure criteria and matching rules for similar instances.

1) Similarity Measure Criteria: The similarity measure criterion refers to determining an appropriate distance metric to calculate the similarity degree between two given HI curve

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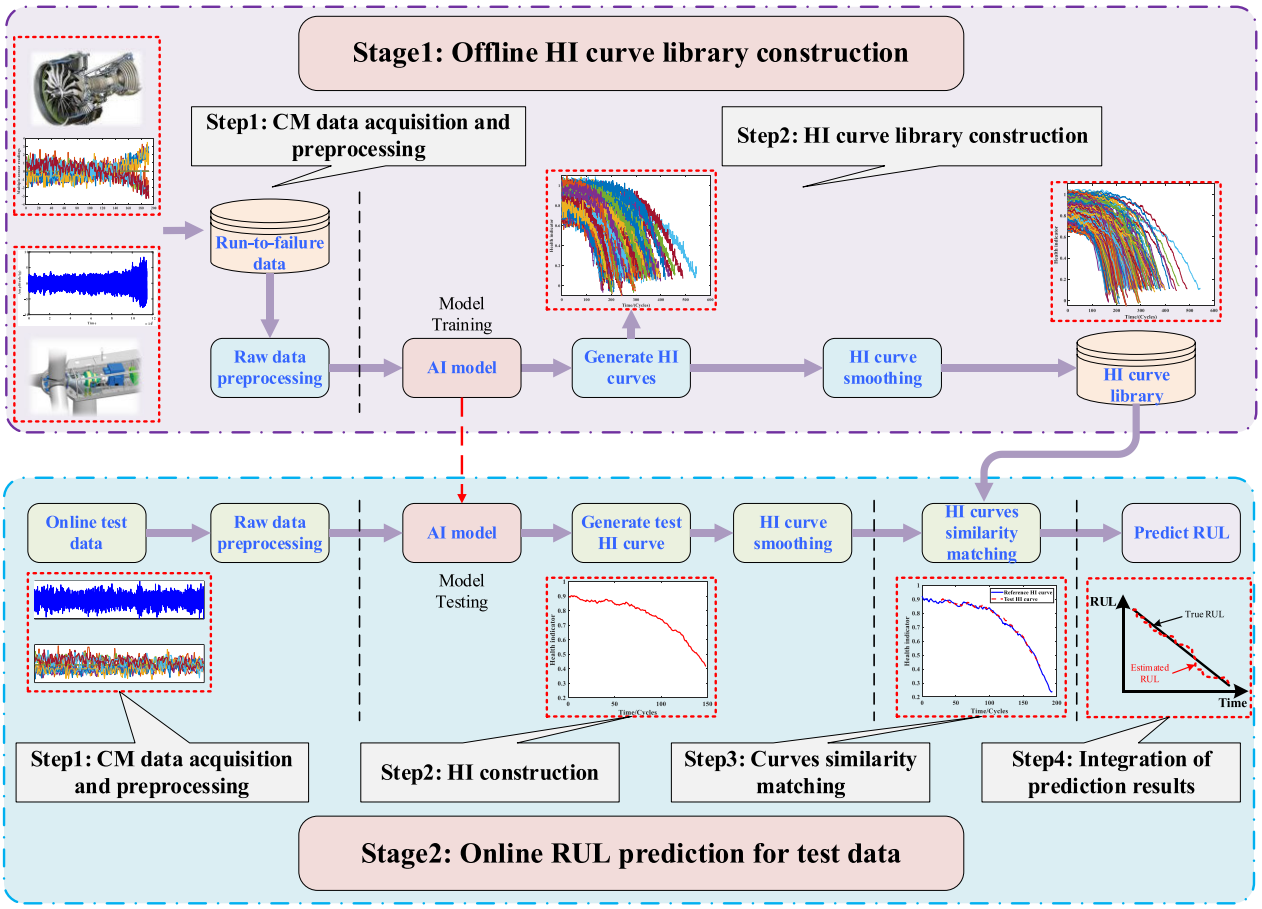


Fig. 3. General procedures of similarity-based RUL prediction methodology.

segments. In early studies of the SBP method, Liu et al. [45] defined the distance between two multidimensional feature vectors using Mahalanobis distance, which was calculated as

$$D_{\mathrm{MD}}(\boldsymbol{H}^j, \boldsymbol{H}^{\mathrm{test}}) = \left(\boldsymbol{H}^j - \boldsymbol{H}^{\mathrm{test}}\right)^{\mathrm{T}} \Sigma^{-1} \left(\boldsymbol{H}^j - \boldsymbol{H}^{\mathrm{test}}\right) \tag{6}$$

where $D_{\mathrm{MD}}(\cdot)$ represents the inverse of the covariance matrix of two eigenvectors. Then, a matching matrix is formed based on the distance function $D_{\mathrm{MD}}(\cdot)$ for the test and reference instances. Each element in the matrix represents the similarity value of the corresponding timestamp. Subsequently, the RUL prediction is achieved by analyzing the matching matrix.

Due to its simplicity and practicability, most researchers define the measurement criteria of curve similarity based on Euclidean distance, including standard Euclidean distance [49], [50], [51], [52], [53], [54], [55], [56], [57], mean Euclidean distance [58], square Euclidean distance [59], and mean square Euclidean distance [60], [61], [62], [63], [64], [65]. A typical distance function for a 1-D HI vector can be expressed as follows:

$$D_{\mathrm{ED}}(\boldsymbol{H}^j, \boldsymbol{H}^{\mathrm{test}}, \tau) = \frac{1}{T_c} \sqrt{\sum_{k=1}^{T_c} \left(h_{k+\tau}^j - h_k^t\right)^2} \tag{7}$$

where τ is the time offset, i.e., the step size of the HI segment translated along the time axis from its initial position. In some studies, for simplicity, the calculated Euclidean distance is directly considered to be the similarity degree between two curve segments. The smaller the distance value, the more similar the degradation process between the test equipment and the reference instance.

Liang et al. [4], [66] provided a more simplified version based on the pointwise difference of trajectories. The mean absolute error (MAE) of all elements is used as a similarity measure method for two HI curve segments with the length of T_c, which can be represented as

$$D_{\mathrm{MAE}}(\boldsymbol{H}^j, \boldsymbol{H}^{\mathrm{test}}, \tau) = \frac{1}{T_c} \sum_{k=1}^{T_c} |\delta(j, k, \tau)|$$
$$\delta(j, k, \tau) = h_{k+\tau}^j - h_k^t. \tag{8}$$

Similar samples with more severe degradation are assigned a greater weight based on the proposed similarity measure, improving the accuracy of the prediction results.

Some scholars have proposed using cosine distance to evaluate the spatial directional similarity and measure the difference between two HI curve segments, whose distance function is defined as

$$D_{\mathrm{CD}}(\boldsymbol{H}^j, \boldsymbol{H}^{\mathrm{test}}, \tau) = 1 - \frac{\sum_{k=1}^{T_c} \left(h_k^t \times h_{k+\tau}^j\right)}{\sqrt{\sum_{k=1}^{T_c} \left(h_k^t\right)^2} \times \sqrt{\sum_{k=1}^{T_c} \left(h_{k+\tau}^j\right)^2}}. \tag{9}$$

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The cosine distance can be independently used as a similarity measure or in combination with other distance functions to comprehensively evaluate the difference between two HI curve segments [4], [50], [67], [68].

Contrary to the traditional distance measurement, Cai et al. [69], [70] measured the difference between test and run-to-failure instances from the data distribution perspective. MMD was used as the metric [71], and multidimensional sensor readings were directly taken as the degradation features. A biased estimate of MMD for samples $\mathbf{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_{n_X}\}$ and $\boldsymbol{Y} = \{\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_{n_Y}\}$ from two data distributions can be expressed as

$$D^2_{\mathrm{MMD}}(\boldsymbol{X}, \boldsymbol{Y}) = \left\| \begin{array}{l} \frac{1}{n_X^2} \sum_{i=1}^{n_X} \sum_{j=1}^{n_X} \kappa(\boldsymbol{x}_i, \boldsymbol{x}_j) - \frac{2}{n_X n_Y} \sum_{i=1}^{n_X} \sum_{j=1}^{n_Y} \kappa(\boldsymbol{x}_i, \boldsymbol{y}_j) \\ \quad + \frac{1}{n_Y^2} \sum_{i=1}^{n_Y} \sum_{j=1}^{n_Y} \kappa(\boldsymbol{y}_i, \boldsymbol{y}_j) \end{array} \right\|_{\mathcal{H}} \tag{10}$$

where $\mathcal{H}$ represents the reproducing kernel Hilbert space (RKHS) and $\kappa(\cdot, \cdot)$ is the kernel function.

Dynamic time warping (DTW) is a timing analysis technique measuring the difference between two time series with different lengths, which can reduce the impact of abnormal samples. In [67], [72], [73], [74], and [75], the DTW distance was used to assess the similarity between in-service equipment and historical failure instances. Some researchers have argued that dividing degradation trajectories into "similar" and "dissimilar" according to predefined thresholds is inappropriate, as similarity measures should be allowed to transition gradually [46], [76]. Therefore, fuzzy similarity is applied to describe the relationship between test trajectories and reference trajectories by resorting to the fuzzy logic modeling paradigm, where the original distance is converted into fuzzy similarity values through membership functions.

Furthermore, unsatisfied with a single distance measurement method, many researchers combine multiple distance measurement methods or mine the similarity between two HI line segments from multiple aspects. Liu et al. [50] simultaneously considered distance and spatial direction similarities, successfully predicting the RUL of slotted tools by combining Euclidean and cosine distance. In order to comprehensively analyze the impact of bearing type and vibration intensity, Wang et al. [77] designed a new difference measurement function combining Euclidean distance and cross correlation similarity. Zeming et al. [78] determined the similarity between degradation trajectories through shape and value differences, considering trajectory pointwise difference, shape factor, and degradation degree measurements. In addition, some researchers designed the measurement index by considering the local and global similarity of degradation trajectories. Wang et al. [79] calculated the first-order difference of HI curve segments as a local degradation trend and developed an information fusion similarity measurement method based on Euclidean distance combined with global similarity. Gu and Ge [68] used weighted Hausdorff distance and cosine similarity to define a compound similarity measurement, which comprehensively considered the local and global similarities.

Researchers also tend to convert the calculated distance into a similarity score function to reflect the similarity more intuitively between two HI curves [60], [63], [64], [67], whose value range is usually limited to [1, 0]. The higher the similarity score, the more similar the two trajectories. A commonly used scoring function can be formulated as follows:

$$\mathrm{Sim}(\boldsymbol{H}^j, \boldsymbol{H}^{\mathrm{test}}, \tau) = \exp\left(\frac{-D(\boldsymbol{H}^j, \boldsymbol{H}^{\mathrm{test}}, \tau)}{\lambda}\right) \tag{11}$$

where $D(\cdot)$ is a well-defined distance function and λ is a relaxation factor controlling the similarity degree for a given distance metric. Francesco et al. [51] provided a similar definition using Gaussian kernel functions, while Zhou et al. [75] provided a simpler form.

2) Matching Rules for Similar Instances: The similar instance matching rule specifies the technical details other than similarity measure criteria in the execution of the similarity matching procedure between in-service and failed instances. More and more researchers are committed to optimizing matching rules to improve the accuracy of degradation pattern matching and the applicability of SBP methods.

Generally, the test and the reference trajectories have different lengths; the former is usually smaller. In most cases, a fixed-length sliding window w is used to crop the reference HI curve to the same length as the test trajectory, thus enabling similarity calculation. The last part of the test HI curve must be included in the cropped segment since the predicted target is the RUL after the last observation moment. Fortunately, performing similarity matching directly on HI curve segments with unequal lengths is still feasible. However, this usually depends on flexible distance metrics, such as DTW and Hausdorff distances [68], [72], [74].

Various strategies can be adopted to determine the time window size. A common approach is setting it to a constant value smaller than the shortest length of all test instances, resulting in different test instances following the same sequence length. Setting the time window w to the minimum length of all HI segments is the simplest solution [64]. However, a satisfactory prediction result may be difficult to achieve. Therefore, researchers generally select appropriate window size values based on operational experience and experiments [50], [65]. The impact of different window lengths on the final prediction performance was discussed in detail in [55] and [72]. Another idea is using the test HI curve as a benchmark, where the time window size varies according to the length of different tested HI curves [67], [80].

Some studies have considered multiple similarity matching paradigms for a single test sample to improve the accuracy of similar instance matching rather than relying solely on a single matching process. Under the influence of unstable operating conditions and unexpected interference, the degradation state of a system may exhibit a different degradation behavior than before. Lyu et al. [72] proposed to segment a single test degradation trajectory into multiple HI segments and conduct multiple local similarities matching to identify the impact of historical events on RUL. However, applying a fixed-length time window to all tested HI curves is considered insufficient

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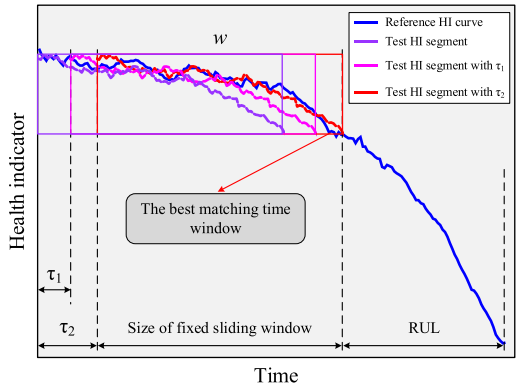


Fig. 4. Schematic of the sequential search method with a time lag τ for finding the best matching reference HI segment.

to utilize observable data and may increase prediction errors. Therefore, Xia et al. [49] used a multiscale time window to process all training and test samples, fusing multiscale similar RULs to estimate the RUL of aircraft engines.

Contrary to using a single HI, Zhou et al. [75] extracted multiple time- and frequency-domain features from vibration signals as damage indicators for similarity calculation. Gu and Chen [81] employed PCA to obtain multiple principal components of degradation indicators for similarity matching. Moreover, the authors predicted the RUL of gyroscopes by weighting the fuse of similar RULs of multiple predictions based on contribution rates. Furthermore, multiple degradation variables were screened in [56] by designing a selection function. Finally, similarity matching was carried out to search for similar reference samples.

Since the initial health of the test instance is usually different from that of the failure instance, the classical best-matched HI curve segment search method is based on the sequential search algorithm that takes a time lag τ into account [48], [53]. In other words, the tested HI segment is translated along the time axis with a certain time step, and the optimal matching segment with the smallest similarity value is found one segment after another, as depicted in Fig. 4. Considering the exponential computational burden of multiple local similarity matching, a coarse-to-fine strategy was designed in [72] to reduce the search space. In order to mine more similar reference instances, Yu et al. [62] proposed a zero-centering rule to allow the HI segment to translate along the HI axis. Specifically, the tested HI segment and the reference HI segment are converted to a zero-mean form within the same time window to focus more on the difference in the shape of degradation curves, which is beneficial for limited reference samples. Considering the errors introduced by abnormal degradation trends, Xue et al. [65] incorporated a degradation consistency test into the SBP method to identify pseudo-similar instances. In addition, the test HI segment was matched with the complete reference degradation trajectory and the windowed reference trajectory to ensure the availability of similar instances [68].

Finally, for the jth reference instance, the inferred reference RUL value of test instance t in its most similar fragment can be calculated as follows:

$$\mathrm{Rul}_{t,j}(\tau, w) = \left(T_j - w - \tau\right) * \Delta T \tag{12}$$

where ΔT represents the sampling interval of two adjacent samples.

C. Curve Similarity Matching

A set of predicted most likely RUL values $\left\{\mathrm{Rul}_{t,j}\right\}_{j=1}^{J}$ and their corresponding highest similarity $\left\{\mathrm{Sim}_{\max}^{t,j}\right\}_{j=1}^{J}$ are obtained once the similarity matching program is completed, which will be used for integrating the final RUL. The remaining life of operating machinery obtained by the SBP method can be presented as point estimation and a probability distribution considering uncertainty quantification.

1) Point Estimation: Point estimation can provide an intuitive prediction result, which is usually obtained by weighted fusion of reference similar RUL values. Since the HI segment with high similarity is considered to have a more likely future degradation trend, its predicted value should be given a greater weight [60]. A classical weighted summation strategy can be expressed as follows:

$$\begin{aligned} \mathrm{RUL_t} &= \sum_j \omega_{t,j} * \mathrm{Rul}_{t,j} \\ \omega_{t,j} &= \frac{\mathrm{Sim}_{\max}^{t,j}}{\sum_j \mathrm{Sim}_{\max}^{t,j}} \end{aligned} \tag{13}$$

where $\omega_{t,j}$ is the similarity weight of the jth reference instance and $\sum_j \omega_{t,j} = 1$. Additionally, Wang et al. [82] also defined weight formulas in the form of a softmax function and an inverse proportional weighting function.

Including all referenced RULs in the final prediction result is considered to reduce the prediction accuracy [55] because the test instances can only obtain smaller $\mathrm{Sim}_{\max}$ values on some reference instances, i.e., outliers. Therefore, outlier elimination is a necessary preprocessing procedure. The 3σ principle is a commonly used method for identifying outliers in engineering, which was adopted by Huang et al. [55]. The K-means clustering algorithm was used in [79] to eliminate matching results with low similarity scores. Most studies control the number of reference instances that participate in aggregation by presetting a threshold coefficient and combining it with the highest $\mathrm{Sim}_{\max}$ value among all matching instances [60], [61]. A more complex culling process was designed in [47], which combined similarity threshold and interval defined by different quantiles to filter abnormal data, removing abnormally long and short RUL values. In addition, Cai et al. [69], [70] applied the kernel two-sample test to screen similar historical samples.

In contrast to the weighted summation strategy, the average of all baseline results was directly calculated in [53] as the final predicted value. Wang et al. [47] only included the upper and lower boundaries of the candidate RULs in the calculation range to obtain a lower error score. Considering the possible degradation difference of similar segments in the future, future bias weights were incorporated into the test system's RUL estimation [72]. Considering the difference in degradation degree, Zeming et al. [78] assigned more weight to reference samples with higher degradation degree to promote reasonable maintenance decisions.

In order to further improve the accuracy and robustness of the prediction results, Yu et al. [62] averaged the

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results of the classical SBP method and the SBP method with the zero-centering rule as the final prediction value. Chen et al. [73] established multiple support vector regression models for RUL prediction and assigned weight coefficients according to the similarity of all components. An ensemble learning model for milling cutter life prediction was developed in [80], using differential evolution SVR and TSBP methods as the base learners. Multiparameter and multisample similarity fusions were combined in [75] to predict the RUL of diesel engine gearboxes. Gu et al. [57] took operating conditions and maintenance into account in the SBP method, where samples with similar operating conditions and effective maintenance would be assigned greater weights.

2) Prediction Interval Estimation: Unlike point estimation, which provides an exact predicted value, the predicted RUL in some studies is presented as a probability distribution, providing a confidence interval and facilitating maintenance decisions.

Kernel density estimation (KDE) is a nonparametric analysis method for estimating unknown probability density functions, which is often combined with SBP methods to provide uncertainty management. Xia et al. [49] used a Gaussian kernel density estimator in KDE to fit the probability density function of the predicted RUL. Moreover, the authors obtained the point estimation of the test engine's RUL via the fit distribution's mean value. Similarly, using KDE technology, the RUL distribution of test instances was also obtained in [72] and [83] based on the reference RUL and the corresponding similarity value.

In addition, many studies have quantified the uncertainty of prediction results by introducing other techniques. Cai et al. [70] obtained a set of similar reference instances using the SBP method. The instances were used to initialize particle states, followed by PF to predict RUL values and provide uncertainty intervals. The SBP method and belief function theory were combined in [84] to provide prediction intervals, demonstrating superior performance to KDE and mean-variance estimation (MVE) methods in the case of a few available reference samples. Yan et al. [64] used the Kolmogorov–Smirnov test to verify the normality of the reference RUL set. Furthermore, the authors formed the probability distribution of RUL for in-service equipment based on historical reference samples. In order to obtain the confidence interval for RUL estimation, Schwartz et al. [52] assigned a 95% confidence interval to each reference degradation trajectory in the HI curve library, resulting in upper and lower bounds for RUL estimation. In addition, the uncertainty was quantified in [69] by fitting the Weibull distribution based on the predicted RULs. Recently, Lin et al. [85] also provided uncertainty in prediction results by linking the SBP method with the Gaussian process.

IV. Time Series Prediction-Based Methods

Different from the SBP method, other HI-dependent prognostic methods require not only the explicitly extracted HI but also the definition of a quantified threshold for the failure state of mechanical equipment. Specifically, both the statistical method and the TSP method for HI fall into this category. Although the determination of FT sacrifices the flexibility of prognostic methods to some extent, it provides good application prospects for solving the prognosis issues of online prediction and limited samples. This section will provide a detailed commentary on these methods, especially the TSP method.

A. Model and Data-Based Statistical Methods

Model and data-based statistical methods achieve machinery RUL prediction by fitting the available observation data into a probabilistic model without relying on physical knowledge or engineering expertise. Moreover, statistics-based approaches typically regard RUL as a random variable and represent it as a conditional probability density function of historical operating data. Therefore, this method has the advantage of providing the RUL uncertainty, which can help the subsequent maintenance decision-making. Since Si et al. [11] and Zhang et al. [12] have reviewed some statistics-based machinery prognostic methods, this section will provide a brief review for completeness and a focus will be placed on recent advances.

As the earliest and simplest prediction models, linear regression [86] and nonlinear regression models (such as polynomial regression [87] and exponential regression [17]) can estimate the relevant parameters based on historical data through the least-squares method or maximum likelihood function; then, the empirical relationship between machinery degradation and state sequences is revealed. The autoregressive (AR) model regards the current state as a polynomial function of historical observations and random factors. Furthermore, the AR model is combined with moving average, PF, or Kalman filter (KF) to predict the machinery RUL [88]. The random coefficient regression model considers the random effect of individuals and adds the random error term subjected to normal distribution into the regression model for machinery prognosis [89]. Moreover, RUL prediction methods based on Markov theory have also been gradually developed due to solid mathematical support [90]. Another method is the proportional hazards model based on covariates. However, these methods do not require a predefined FT, so they are not presented in detail. The stochastic process model has unique advantages in describing the randomness of machinery degradation. Hence, it is widely used for the RUL prediction of RM. Since the Wiener process (WP) is a Brownian motion with a drift term that can easily provide the predicted probability distribution. Therefore, it is widely applied to risk analysis and maintenance decision-making [13], [91]. In [12], machinery prognosis methods based on WP were systematically summarized. Moreover, Zhang et al. [92] presented some recent advances in the application of WP to the prognosis field of mechanical equipment. The gamma process (GP) model assumes that machinery deterioration occurs gradually over time in a tiny positive increment following the gamma distribution, and the increments in different time intervals are independent of each other [93]. RUL prediction methods for different mechanical equipment based on GP models were developed in [94] and [95]. Contrary to the GP model, the inverse Gaussian process (IGP) model assumes that the degradation increment

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follows the inverse Gaussian distribution. It has also been used to model the machinery degradation process and predict its RUL [96], [97].

The statistical model-based RUL prediction method can easily achieve higher prediction accuracy under a certain condition. However, with the increasing complexity and scale of the RM, its degradation process is becoming more complex and variable. Additionally, these statistical models rely on some particular assumptions, such as normal distribution and linear relationships, and deviations from these assumptions in actual data can significantly impair the model's efficacy. Due to the low flexibility, large demand for data, and heavy computing burden of the statistical model-based prognostic methods, they are gradually replaced by AI-based methods in the intelligent manufacturing era.

B. Intelligent Time Series Prediction Methods Based on HIs

The method based on HI TSP is another unique machinery prognostic scheme. The entire forecasting procedure of the TSP method is shown in Fig. 5. The TSP method first converts the raw CM data into HI to characterize the degradation trend of RM through SP techniques, manifold learning algorithms, or neural networks. Then, based on the historical observations or known HI values, ML or DL models are utilized for predicting the future health status in single or multistep ahead. This process is repeated until the predicted HI value exceeds a predetermined FT [98]. Then, the monitored mechanical component is evaluated to have reached its end of life (EOF). Finally, the corresponding RUL value can be calculated by multiplying the number of HI values predicted before exceeding the threshold by the sensor sampling interval. The above process demonstrated that this method relies on extrapolating the machinery's health status and does not require RUL labels in the prediction process. In addition, unlike end-to-end pattern recognition and SBP methods, the TSP methods are based on individual inference, while they are group-based reasoning. Therefore, one of the characteristics of TSP methods is that they are very suitable for RM prognosis issues with limited samples.

Similar to the SBP method, the TSP method also requires explicit HI acquisition from historical degradation data. Furthermore, its prediction accuracy highly depends on the monotonicity and robustness of the constructed HI. Since various HI construction methods have been summarized in Section II, the following will focus on the ahead prediction strategies, determination principles of FT, and advanced TSP models in TSP methods. In addition, filter-based RUL prediction methods (such as PF) have many commonalities with TSP methods, and there are limited publications on narrow TSP research. Therefore, some filter-based machinery prognostic methods are also included in the following discussion.

1) Ahead Prediction Strategies: Generally, the commonly used ahead prediction strategies in time series forecasting can be divided into one-step and multistep strategies [99], [100], [101]. In one-step ahead prediction, only a single point is obtained in each iteration, while a series of points can be obtained in multistep ahead prediction, as shown in Fig. 6. In this figure, the initial prediction time means the time instant at which RUL prediction is started. It can be usually determined using the triple standard deviation principle. The first arrival time represents the time instant at which the predicted HI first exceeds the FT. It can be calculated by the following formula:

$$F_t = I_t + N_{\mathrm{HI}} \times (T_d + T_s) \tag{14}$$

where I_t is the initial prediction time, N_{HI} is the number of predicted HI values, T_d is the sampling duration, and T_s is the interval time. Statistically, most studies tend to estimate the RUL of machinery by extrapolating its health status through single-step advance prediction.

Yan et al. [102] selected the relative rms (RRMS) value of vibration signals as the HI and obtained the RUL of the rolling bearing by progressively predicting future RRMS values through a polynomial regression model. In order to monitor the health status of helicopter planetary gear plates and predict the RUL, Chen et al. [103] calculated the sideband ratio as an HI. Then, based on the adaptive neuro-fuzzy inference systems and the higher order PF, the future health state is predicted in p-step ahead to obtain the RUL value of the monitoring system. Cheng et al. [104] adopted a similar method for RUL estimation of wind turbine gearbox bearings and proposed an enhanced PF algorithm to solve the particle impoverishment problem during resampling. Peng et al. [105] also conducted a similar study, constructing HI through DBN and incorporating a resampling smoothing method based on a fuzzy inference system into the PF algorithm to avoid the particle degeneracy issue. Zhang et al. [106] used BiLSTM as the prediction model to track HI degradation fused by a single-layer perceptron and achieved RUL prediction of turbofan engines by performing single-step ahead prediction. In addition, Lu et al. [99] focused on the error superposition problem in time series forecasting. The authors proposed the idea of adversarial learning to reduce the cumulative error caused by multiple iterations of one-step prediction.

However, single-step-ahead prediction is considered to be time-consuming in some cases. Hence, some researchers have actively explored direct multistep ahead prediction. For example, Xiao et al. [100] utilized wavelet packet analysis to extract the energy as degradation indicators and then used a backpropagation neural network to perform rolling multistep ahead prediction and obtain the RULs of rolling bearings. Furthermore, the authors established an adaptive time window that can be adjusted according to the increasing rate and fluctuation factor. Subsequently, Kong and Li [107] used a similar time series forecasting strategy to carry out a multistep advance prediction for rms values of tool vibration signals using stacked BiLSTM until the predicted value exceeded the given FT. Moreover, Li et al. [101] considered the processing method containing outliers in the adaptive time window, further improving the accuracy of the proposed method. Based on the above research, the identification method for the first prediction time was further explored in [108] by constructing a stacked BiLSTM model for future HI prediction and obtaining the RUL of rolling bearings. However, a common drawback of these methods is that only a single training sample is included

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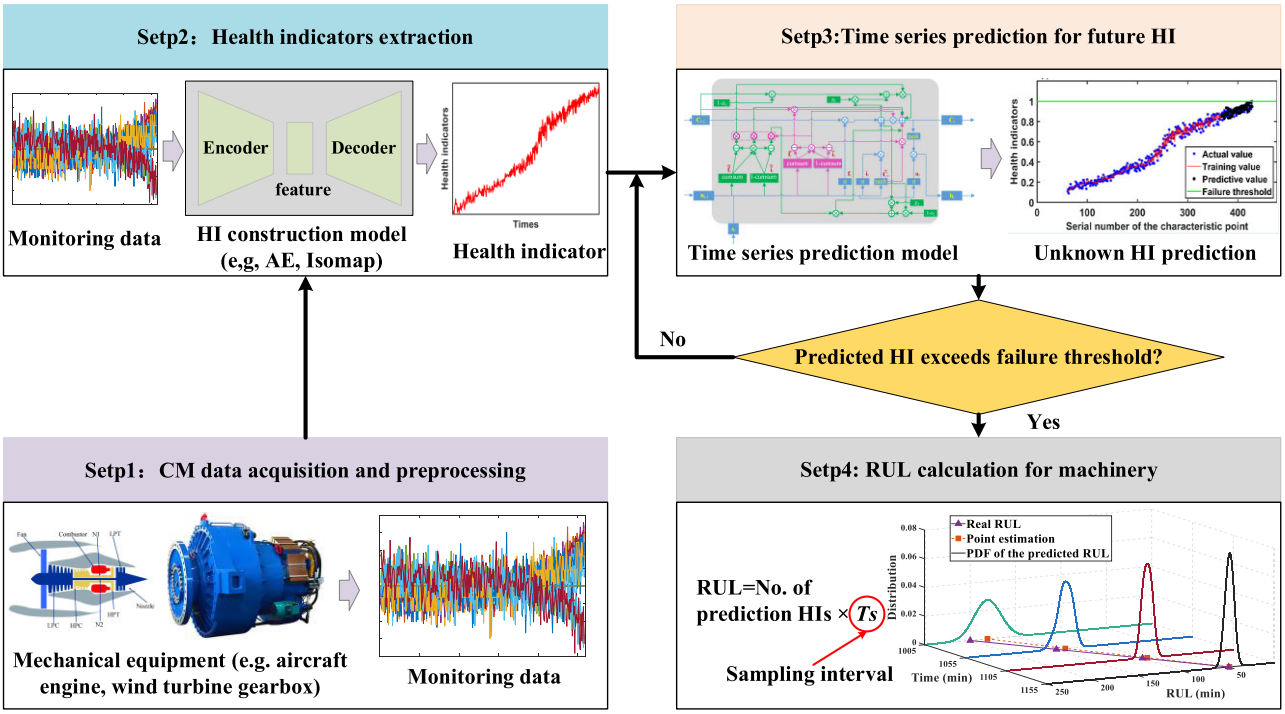


Fig. 5. Flowchart of the RM prognostic methodology based on time series prediction of HI.

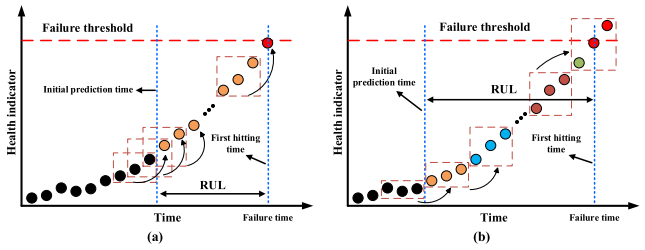


Fig. 6. Two different ahead prediction strategies. (a) Single-step ahead prediction and (b) multistep ahead prediction.

in prediction network training, and insufficient training samples may decrease the prediction performance. Additionally, the way FTs are determined has not yet received attention.

2) Failure Threshold Determination Principles: Setting corresponding FTs for various RM is an open problem [109]. However, this can be tricky because different individuals' failure types and degradation rates may differ even for the same rotating part. Moreover, different devices have varying tolerance levels for component failures, and the definition of failure states is unclear, further increasing the difficulty of setting a unified FT. Consequently, researchers mainly set the appropriate FT by taking the mean value, making empirical assessments, setting up experiments, or constructing a good HI.

Li et al. [110] proposed an enhanced metabolism gray forecasting model combined with PF to address the challenge of failure data scarcity, where the FT was set to the average value of monitoring indicators when the pump was forced to shut down. Chen et al. [73] calculated the mean value of the last five local HI points as FT. According to the bearing degradation trend, Qin et al. [111] set the maximum rms value as its FT. The moment when the rms value significantly jumps was empirically identified as the failure time point for rolling bearings in [112], [113], and [114]. He et al. [115] considered FT as an experimental parameter and set the FT of the bearing as the HI amplitude corresponding to when its RRMS increased sharply and exceeded a certain value. Yang et al. [116] constructed the HI based on self-normalizing CNN and GRU to avoid the influence of man-made factors such as expert knowledge and work experience on FT, where the value of one was determined as a unified FT. The final RUL was inferred from a double exponential degradation model and an unscented KF algorithm. On the other hand, Zhou et al. [117] proposed a distribution contact ratio metric for unsupervised HI construction, contributing to a uniform FT.

In addition to determining FT from the perspective of the HI value, some research work also addresses the challenge of setting FT from the methodological perspective. As early as 2014, Javed et al. [21] explored a procedure for dynamic FT assignment using subtractive-maximum entropy fuzzy clustering (S-MEFC). Within this procedure, the unsupervised clustering algorithm allows each classifier to have different degraded states. The machine's RUL can be estimated by iterative one-step ahead prediction using the summation wavelet-extreme learning machine (SW-ELM) until its discrete state intersects the FT. Subsequently, Zhou et al. [118] conducted a similar study combining time series forecasting and discrete state division. The main difference is that the reduced kernel recursive least-squares (RKRLS) algorithm was used for predicting degradation features, and the FT was no longer dynamically allocated. Based on similar ideas, Wang et al. [119] utilized LSTM as a sequence-to-sequence

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regressor to perform single-step ahead prediction. The authors employed a deep forest classifier (DFC) to complete health state estimation. The core idea of the above study is integrating a prediction model and a classification model to simultaneously perform HI forecast and discrete state estimation, thereby achieving RUL prediction without explicitly presetting FTs.

3) Advanced Time Series Forecasting Models in the TSP Method: Apart from determining the ahead prediction step size and FT, a powerful TSP model is also crucial to the TSP method's final result. Accordingly, some recent studies have focused on improving the information mining capability and long-term prediction ability of time series forecasting models.

a) RNN-based prediction methods: A self-calibration temporal convolutional network (SCTCN) was proposed in [115] to enhance the local information and avoid long-term dependencies. In addition, the impact of different time window sizes on prediction results was also discussed. Due to the fact that the sequence prediction for HI is essentially a time series modeling problem, the RNN-based prediction network structure has been adopted and improved by most researchers. By replacing the fully connected in LSTM with a convolutional operator and embedding a self-attention module, Li et al. [112] strengthened the nonlinear modeling capability of the prediction model and suppressed useless data. Xiang et al. [113] enhanced the network's ability to mine the key degradation information by embedding a multihead attention module into BiLSTM and using it for the RUL prediction of rolling bearings. In addition, Qin et al. [40] proposed a novel macroscopic–microscopic attention (MMA). They incorporated it into LSTM to pay attention to the contributions of different input data, thus improving the prediction accuracy of the gear's RUL. Yan et al. [120] adopted ordered neuron LSTM (ON-LSTM) to iteratively predict the frequency center index of gears, effectively utilizing the sequential information among neurons. The LSTM with attention-guided ordered neurons (LSTM-AON) and cocktail LSTM (C-LSTM) neural networks were further developed in [121] based on the ON-LSTM to enhance the hierarchical division ability of degradation information and improve the network's long-term prediction capability.

Apart from LSTM, some studies have also been implemented to improve the predictive performance of GRU networks. A gated dual attention unit (GDAU) was constructed in [111] by designing two attention gates at GRU's input and output sides to improve the accuracy of bearing RUL prediction and speed up the convergence speed of the network. Additionally, Chen et al. [73] further considered the adaptive partitioning of the sequence information for attention gates and then proposed the gated adaptive hierarchical attention unit (GAHAU) for predicting the RUL of servo motors successfully. On the other hand, Zhou et al. [122] focused on strengthening the forgotten historical information and proposed a reinforced memory GRU (RMGRU) for machinery prognostics. On this basis, a dual-threaded GRU (DTGRU) was developed [123] to extract the long-term degradation trend and local input signal fluctuations, enhancing its predictive ability for time series.

b) Transformer-based prediction methods: Transformer network is a groundbreaking DL model that has revolutionized natural language processing tasks. Introduced in the article "Attention is All You Need" by Vaswani et al. [124], the transformer architecture relies solely on self-attention mechanisms without recurrent or convolutional layers. This attention mechanism allows the model to weigh the importance of different input sequences when performing prediction. Transformers excel in capturing long-range dependencies in sequential data, making them highly effective for processing complex time series. As a result, the transformer and its improved versions have been applied to machinery prognostic tasks. Fu et al. [125] utilized a nonstationary transformer to predict the trend of HI, which successfully mitigates the influence of nonstationary information on the prediction results. In order to enhance the capacity of the model to extract more discriminative time correlation features, Zhang et al. [126] proposed a weighted time embedding transformer network for bearing RUL prediction by imposing adaptive weight constraints on the transformer's inputs. Guo et al. [7] proposed a multiscale hourglass-transformer based on 1-D CNN, which can scale the time sequence into multitime scales for feature fusion. For enhancing the prediction accuracy, Zhou et al. [127] developed a new transformer network by embedding a feature fusion gate, which could dynamically fuse global and local features. Ren et al. [128] believed that time series with different lengths have their characteristics, and proposed a dynamic length transformer to adaptively learn characteristics of time series with different lengths for machinery RUL prediction. While transformer and its variants have demonstrated remarkable performance, their reliance on self-attention mechanisms and the necessity to process entire sequences require substantial computational resources and memory. This brings the issues of scalability and efficiency, especially when processing the extensive time series data or large datasets. Additionally, transformer heavily depends on self-attention, which can be affected by outliers or irrelevant features within the sequence, potentially resulting in subpar prediction.

c) Graph neural network-based prediction methods: Graph neural networks (GNNs) are being increasingly applied to RUL prediction, exhibiting their effectiveness in modeling intricate relationships and dependencies hidden in the monitoring data. Usually, the system components and their interactions are represented as a graph, where nodes correspond to components and edges represent relationships between them. As a result, GNNs can effectively capture the evolving laws and degradation patterns of mechanical systems over time. The ability of GNNs to model the dynamic relationships between components allows for more accurate and robust RUL prediction compared to traditional methods. By introducing autoregressive moving average and the physics equations of balancing economy and security, He et al. [129] proposed a physics-informed GNN to fuse multisensor data. By replacing the original linear connection in GRU with GCN, Wang et al. [130] developed a gated GCN for machinery prognosis of multisource sensing signals and obtained its confidence interval using quantile regression (QR). Xiao et al. [131] proposed a multiplex aggregation

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heterogeneous GNN to extract the heterogeneous features of the graph as well as the temporal dependency of each node. To effectively extract the hidden spatial correlations along the local and global time axes and heterogeneities, Zhang et al. [132] designed a spatial–temporal dual-channel adaptive graph convolutional network based on the learnable local and global graphs for the turbofan engine RUL prediction. Zhou and Wang [133] proposed a multiperspective spatial–temporal graph attention network to capture the complex global and local spatial–temporal dependencies in multisensor data. With regard to GNN-based methods, creating a suitable graph framework to describe the interconnections among these signals is a challenging task. Establishing the connectivity and assigning appropriate edge weights within the graph are important for capturing the relevant spatial relationships. Furthermore, while GNNs are good at capturing the spatial relationships, they may encounter difficulties in modeling the prolonged temporal dependencies, which are crucial for accurately predicting the degradation tendency over a long period.

4) Prediction Interval Estimation of TSP Methods: To address the issue that intelligent TSP methods typically yield point estimates for prediction outcomes, a variety of techniques have been developed to provide prediction interval estimates for the RUL. Researchers have integrated QR loss with TSP models to offer predicted confidence intervals by estimating various quantiles, such as the 10th and 90th percentiles. Xu et al. [134] employed QR loss to obtain the probabilistic RUL prediction results with uncertainty expression. Zhang and Wang [135] fused QR loss with LSTM networks to construct a temporal quantile regression network. Ly et al. [136] introduced a multiple noncrossing QR loss to adapt T-shaped data, enhancing the precision of prediction intervals. Additionally, dropout not only effectively prevents overfitting but can also be used for interval prediction. By activating dropout during the testing phase, a suite of prediction models is generated and their predicted results are amalgamated to form a predictive distribution. Chen et al. [137] and Zhou et al. [138] adopted this strategy to quantify the uncertainty of predicted RUL and to compute prediction intervals. Bayesian neural networks introduce Bayesian statistical principles into neural networks by assigning probability distributions to the network's weights and biases, thereby enabling the quantification of model uncertainty. Mazaev et al. [139] extended Bayesian neural networks to Bayesian CNN to enhance the predictive performance of RUL with uncertainty assessments. Zhang et al. [140] proposed the Bayesian mixture neural network to achieve more precise RUL predictions and to offer uncertainty estimates. QR loss and dropout technique have simple implementation schemes and good robustness to outliers, but the interpretability of their predictions faces challenges. The theory of Bayesian neural networks is relatively complete, but it requires considerable computational resources, and applying it to large datasets is quite difficult.

V. Current Challenges and Prospects

According to the above analysis and summary, the RUL prediction technology with HI dependence for RM has made many exciting progresses. However, the machinery prognosis in practical industrial environments still has some issues that need to be further explored in the future. In this article, the current challenges and prospects are presented as follows.

- How to increase the interpretability of HI?

It is valuable for constructing the interpretable HIs. While DL models offer powerful feature representation capabilities, their "black-box" nature hinders the understanding of their decision-making processes. This is particularly important for domain experts to trust and use the constructed HI. The interpretability of HI can be studied through three key approaches.

1) *Feature Importance Analysis:* Investigating the relative importance of input features can provide insights into the model's decision-making and help identify the most influential factors contributing to degradation.

2) *Integration of Domain Expertise:* Expert knowledge and domain-specific information are incorporated into the HI construction model design and feature selection process.

3) *Mechanism-Oriented HI Construction:* Combining physics-based and data-driven models offers a powerful approach.

The physics-based model elucidates the underlying degradation mechanisms, while the data-driven model learns the specific degradation patterns from available data. By integrating the physical degradation mechanism into the objective function for HI construction, a more explainable and accurate HI can be built.

- How to predict the RUL of critical components for mechanical equipment operating under time-varying operating conditions?

Most publications mainly focus on RUL prediction methods for RM under a single operating condition. However, in practical industrial scenarios, mechanical equipment inevitably operates under time-varying operating conditions, such as industrial robots and wind turbines, owing to the changes in operational tasks and wind environments. In these cases, the significant fluctuations in speed and load can make critical components in rotating equipment more vulnerable to failure. More seriously, the time-varying operating conditions cause variation in vibration amplitude or rotational frequency, making the accurate estimation of the equipment's RUL even more challenging. Researchers may consider the following aspects to address this challenge: 1) develop some advanced SP techniques that can handle the nonstationary signals and effectively capture the degradation patterns under time-varying operating conditions; 2) design AI models that can reduce the temporal dependencies between the operating conditions and component degradation; and 3) integrating dynamic approaches with DL techniques to construct digital twin models for the RUL prediction of equipment.

- How to deal with the explainability issues of DL models in machinery RUL prediction?

Explainability can be defined as information that helps users understand the model's behavior and the process it undergoes in decision-making. Although DL-based methods have achieved state-of-the-art predictive performance in the field of machinery prognosis, the "black-box" attribute hinders their

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promotion and application in practical industry. Researching the interpretability of RUL prediction models helps guide engineers in building high-performance models and making persuasive decisions. Currently, the preliminary research on this topic has appeared in [141] and [142]. Moreover, exploring the model-agnostic interpretable methods for machinery prognostics may be an available solution. Techniques like SHapley Additive exPlanation or rule-based models can be applied to DL models for providing local or global explanations. These methods simulate the behaviors of DL models and provide insights into the important features or patterns contributing to the RUL prediction.

- How to improve the generalization ability of RUL prediction model?

Predicting the RUL of rms operating under different conditions or facing various production tasks often requires customized training of multiple prognostic models, which results in a waste of time and storage resources. Improving the generalization ability of DL models in various scenarios has become another challenge in machinery prognosis. First, exploring the integration of failure mechanisms of rotating components in RUL prediction is a feasible approach. For example, constructing a digital twin model that combines these failure mechanisms with monitoring data can facilitate the generation of high-quality data. Moreover, the application of domain knowledge can aid in designing more rational feature extraction methods and model structures. Second, the tremendous success of ChatGPT provides another feasible direction. The development of large model technology combined with reinforcement learning requires additional efforts in the future to stimulate the additional technological innovation in RUL prediction tasks. To address this challenge, apart from data augmentation and general large model techniques, regularization, ensemble learning, and pretraining techniques are also worth exploring.

- How to promote the development of online RUL prediction techniques of machineries in real industry?

Although RUL prediction methods have been developed for various rms values, there is still a lack of mature industrial cases on the online prediction. Two critical aspects may hinder the development of this field. First, the lightweight technology is crucial for real-time requirement and rapid response in practical applications. Lightweight models are easier to deploy on edge devices with the limited computing resources. Second, the time-varying operating conditions and external environment interference put forward higher requirements for the robustness and accuracy of online RUL prediction systems. In this context, tiny ML (TinyML) emerges as a promising solution to promote the development of online RUL prediction techniques in real industry. Achieving this goal can utilize techniques like model pruning to simplify the model by eliminating insignificant connections or parameters, as well as employ knowledge distillation methods to train smaller models for replicating the outputs of larger models. By leveraging TensorFlow Lite, the lightweight and efficient ML models can be deployed on low-power consumption edge devices for machine fault prognostics. Furthermore, the robustness and accuracy of online RUL prediction systems can be enhanced by utilizing such techniques as transfer learning, ensemble learning, and anomaly detection.

- How to quantify the uncertainty in machinery RUL prediction?

Since machinery prognosis deals with the forecasting of future behavior, many uncertain factors can affect the prediction results and the ultimate real outcome, such as measurement uncertainty and model uncertainty. On the one hand, the appropriate quantification and processing of the uncertainty in RUL prediction can provide more accurate and comprehensive prediction results. On the other hand, uncertainty management of prediction results can provide a reference basis for the subsequent maintenance decision. The statistics and Bayesian theory-based methods can provide the estimation of uncertainty. Unfortunately, some HI-dependent prognostic methods can only provide confidence intervals through multiple predictions, whereas the estimated confidence intervals are not reliable in real-world applications, as their uncertainty factors come from the model's random initialization. Therefore, developing advanced deep probabilistic models specifically tailored for RUL prediction may be a promising direction. These models can capture the complex dependencies and uncertainty in the sensor data, enabling more accurate and reliable uncertainty estimation.

VI. Conclusion

The manufacturing industry is constantly developing toward precision and intelligence in the tidal wave of the Fourth Industrial Revolution. The PdM technology centered on machinery health CM and RUL prediction is increasingly valued in the intelligent operation and maintenance of high-end equipment. This article presents a comprehensive review of HI-dependent RM prognosis methods based on sufficient literature investigation. First, a key preliminary preparation before performing RUL prediction, namely CM data acquisition, was briefly described. Subsequently, the general procedures and pivotal elements of three machinery prognostic methods exhibiting strong HI dependence, namely, the SBP method, the statistical method, and the TSP one, were mainly discussed and analyzed. Finally, some current challenges and future research directions were pointed out. It can be concluded from the existing research that the three types of HI-dependent prognostic methods differ in their applicability due to their intrinsic characteristics. In general, the SBP method is the best choice in scenarios where there are many run-to-failure samples, but the FT is difficult to determine. On the contrary, when there are few life-cycle samples and a definite FT exists, the TSP method has the potential to achieve satisfactory prediction accuracy. Certainly, the three methods mentioned above can all serve as the alternative option when the life-cycle samples are sufficient and the FT is definite. Moreover, we find that the current HI-based prognostic methods lack close integration with machinery health monitoring (e.g., anomaly detection, health stage division, and fault warning), failing to fully exploit the potential value of HI, and thus, this research direction is worth exploring.

It should be noted that it is impractical to cover all publications relevant to the scope of this article. Nevertheless,

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valuable guidance is provided for engineers and researchers in the related industry and academia, promoting the flourishing of machinery prognosis.

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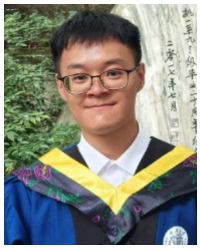
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